



NGUYEN NHAT PHAT

THIRD-YEAR STUDENT

Artificial Intelligence—Computer Vision Concentration

PROFILE

Motivated third-year AI student at HCMC University of Technology and Education with a strong foundation in programming (Python, C++, C#) and computer vision. Known for excellent problem-solving skills, attention to detail, and effective teamwork on complex projects. Passionate about leveraging artificial intelligence and machine learning to develop innovative solutions to real-world challenges. Continuously eager to learn new technologies and contribute creative ideas in collaborative environments.

PROJECTS

- Cassava Leaf Disease Classification with MobileNetV3 (Oct 2024 - Dec 2024):** Developed a convolutional neural network (CNN) using MobileNetV3 to classify cassava leaf diseases. Implemented data preprocessing, augmentation, and k-fold cross-validation for robust model evaluation. Created a graphical user interface (GUI) for easy interaction with the trained model.
- Adaptive Density Pruning for Efficient 3D Gaussian Splatting (2026):** Built a geometry-aware pruning method for 3D scene reconstruction models that accumulate millions of redundant primitives during training, causing excessive memory usage. Introduced adaptive density estimation to detect and remove spatially redundant components while preserving critical structures — achieving 30% smaller model size with no quality loss, outperforming all prior compression methods — validated on 11 scenes across two standard benchmarks.
- Advanced Retinal Vessel Analysis Using Deep Learning for High-Resolution Image Segmentation (Research Project, 2024):** Conducted a deep-learning research project to segment retinal blood vessels from medical images. Applied image processing techniques and convolutional neural networks to achieve accurate segmentation. Received a “Good” evaluation in the university research program.
- Additional Projects:** Explore other projects on my GitHub profile.

TECHNICAL SKILLS

- Languages:** Python, C++, C#, SQL.
- ML / DL Frameworks:** PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, OpenCV.
- Tools & MLOps:** Git, GitHub, Docker, Kaggle, Colab.

CERTIFICATES & AWARDS

- IELTS 6.0 (English proficiency certification)
- Talented Student Scholarship (HCMUTE)
- Student of Five Merits—Ho Chi Minh City Level
- Additional honors and achievements listed on my LinkedIn profile

EXTRACURRICULAR ACTIVITIES

- Volunteer Work:** Participated in community initiatives such as Spring Volunteer campaigns, exam season support programs, and charity/community runs.
- International Academic Exchange:** BMW Program - Digital Twin & AI-Based Industrial Simulation Inbound Academic Exchange, University of Ulsan, South Korea (Jan 2026).

CONTACT

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EDUCATION

2023 - NOW
HCMC UNIVERSITY OF
TECHNOLOGY AND EDUCATION

- Information Technology
- GPA: 3.63 / 4.0

SOFT SKILLS

- Teamwork
- Problem-solving
- Adaptability
- Leadership
- Time Management
- Critical Thinking

RESEARCH INTERESTS

- 3D Reconstruction
- Medical Image Analysis
- Deep Learning
- Object Detection

LANGUAGES

- Vietnamese (Fluent)
- English (Quite Fluent)